Assignment - 03 Oxide Isolation

Isolation in a device is required for providing electrical isolation, minimizing leakage current, and offering structural support. It's a fundamental building block that ensures the smooth operation of complex electronic devices.

This assignment explores transistor design by investigating two key isolation techniques: LOCOS and STI. We'll analyze the impact of oxide thicknesses and gate pitch on LOCOS performance. For STI, dishing in silicon oxide after CMP and pad oxide etch selectivity will be examined. We'll delve into a real-world 14nm logic process, identifying the isolation technique and their rationale.

Question 1: LOCOS

The first task for this assignment is to identify isolation oxide, gate oxide, gate length, gate pitch and bird's beak during LOCOS process. Also, we need to see the change in achieved gate length with respect to planned gate length by plotting a graph.

Local Oxidation of Silicon (LOCOS) is a traditional technique to create isolation regions between transistors and other devices on a silicon wafer. It essentially grows a thick oxide layer specific areas in Silicon wafer, electrically isolating them from the surrounding silicon.

Figure 1 shows the LOCOS process as given in the reference.

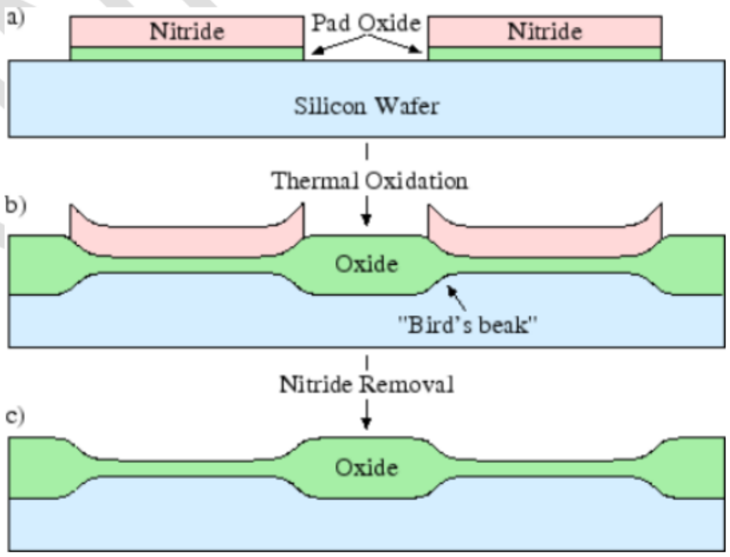

Figure 1 : LOCOS Process Flow

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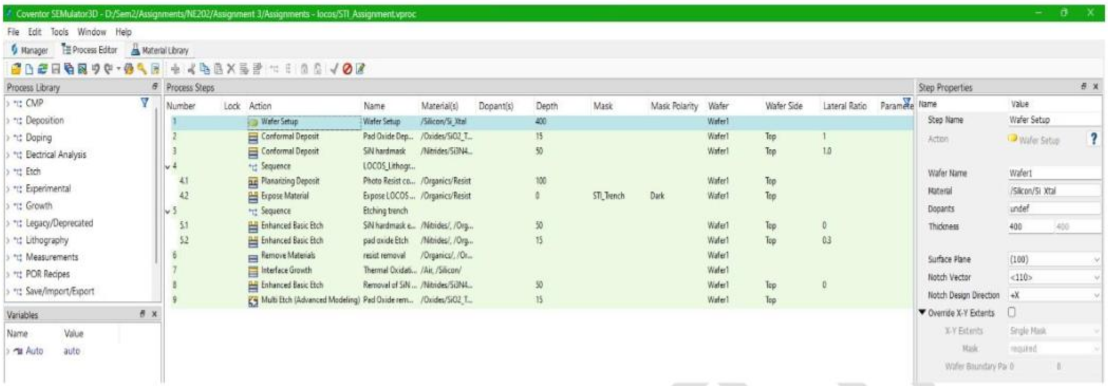
In the above image in second step we can clearly see formation of bird's beak that is thelift in nitrite layer due to excess oxidation on the corners.

Figure 2 to Figure 4 shows the LOCOS process steps in SEMulator3D.



Coventor SEMulator3D - D:/Sem2/Assignments/NE202/Assignment 3/Assignments - locos/STI_Assignment.vproc

Process Steps

Number	Lock	Action	Name	Material(s)	Dopant(s)	Depth	Mask	Mask Polarity	Wafer	Wafer Side	Lateral Ratio	Paramete
1		Wafer Setup	Wafer Setup	/Silicon/Si_Xtal		400			Wafer1			
2		Conformal Deposit	Pad Oxide Dep...	/Oxides/SiO2_T...		15			Wafer1	Top	1	
3		Conformal Deposit	SiN hardmask	/Nitrides/Si3N4...		50			Wafer1	Top	1.0	
4		Sequence	LOCOS_Lithogr...									
4.1		Planarizing Deposit	Photo Resist co...	/Organics/Resist		100			Wafer1	Top		
4.2		Expose Material	Expose LOCOS...	/Organics/Resist		0	STI_Trench	Dark	Wafer1	Top		
5		Sequence	Etching trench									
5.1		Enhanced Basic Etch	SiN hardmask e...	/Nitrides/, /Org...		50			Wafer1	Top	0	
5.2		Enhanced Basic Etch	pad oxide Etch	/Nitrides/, /Org...		15			Wafer1	Top	0.3	
6		Remove Materials	resist removal	/Organics/, /Or...					Wafer1			
7		Interface Growth	Thermal Oxidati...	/Air, /Silicon/					Wafer1			
8		Enhanced Basic Etch	Removal of SiN ...	/Nitrides/Si3N4...		50			Wafer1	Top	0	
9		Multi Etch (Advanced Modeling)	Pad Oxide rem...	/Oxides/SiO2_T...		15			Wafer1	Top		

Step Properties

Name	Value
Step Name	Wafer Setup
Action	Wafer Setup
Wafer Name	Wafer1
Material	/Silicon/Si Xtal
Dopants	undef
Thickness	400
Surface Plane	(100)
Notch Vector	<110>
Notch Design Direction	+X
Override X-Y Extents	☐
X-Y Extents	Single Mask
Mask	required
Wafer Boundary Pa	0

File
Edit
Tools
Window
Help
Manager
Process Editor
Material Library
Process Library
CMP
Deposition
Doping
Electrical Analysis
Etch
Experimental
Growth
Legacy/Deprecated
Lithography
Measurements
POR Recipes
Save/Import/Export
Variables
Name
Value
Auto
auto

Figure 2 : Steps involved in LOCOS Process (Process Editor)

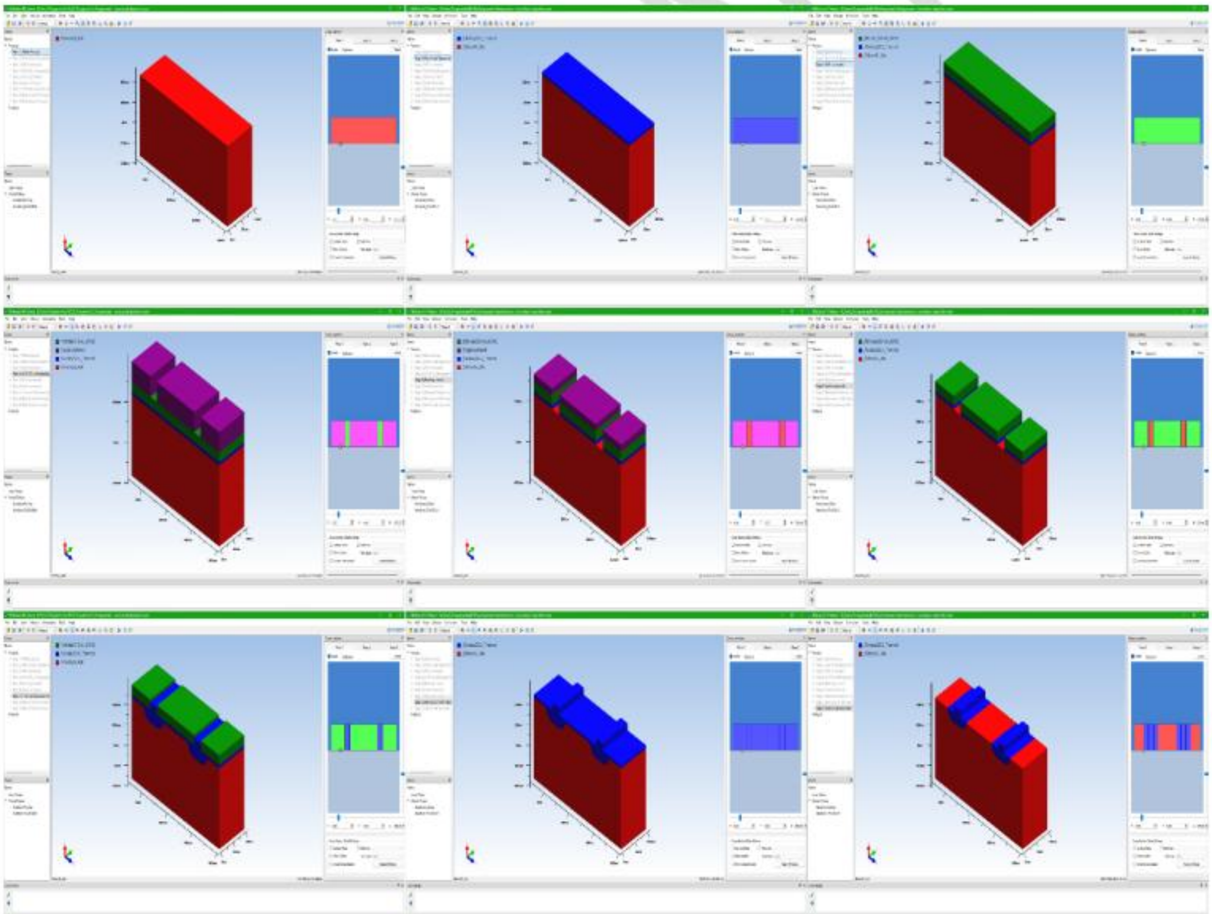


Figure 3 : Simulated LOCOS Process (3D view)

Table 1 : Thickness Specification

S.No.	**Parameter**	**Desired Value (nm)**	**Measured Value (nm)**
01.	Wafer	400	
02.	Gate Oxide (Pad Oxide)	15	
03.	Isolation Oxide	100	
04.	Gate Length	300	
05.	Gate Pitch	350	
06.	Isolation Length	50*2	

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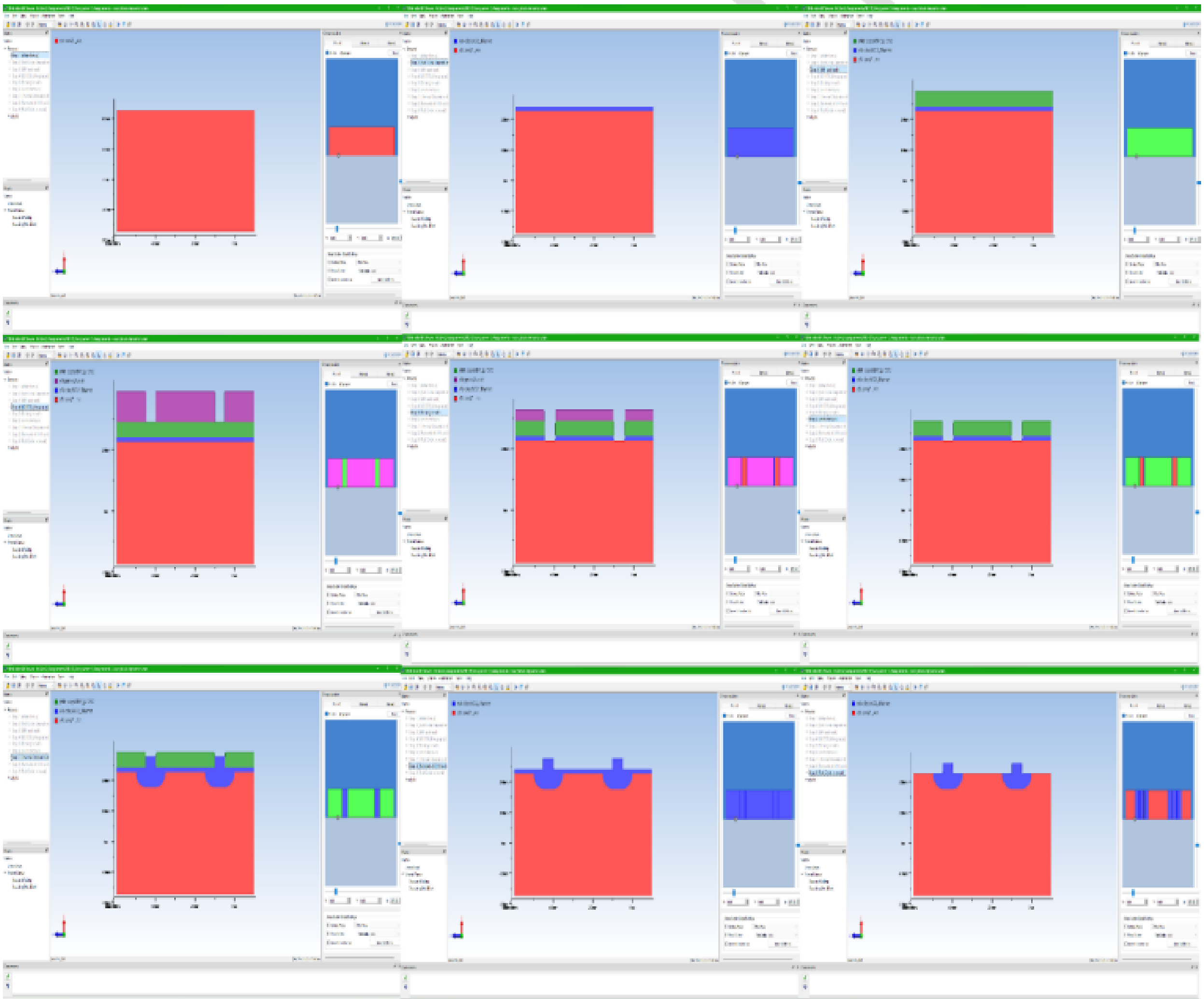

Figure 4 : Simulated LOCOS Process (Front view)

Gate oxide is pad oxide deposited before nitride layer on the region above active area of the device while isolation oxide is a thick layer of thermally grown oxide which provides insulation between two devices.

Figure 5 identifies isolation oxide and gate oxide. oxide deposited in trenches is marked as isolation oxide whereas the pad oxide between active area and nitrite layer is called as gate oxide.

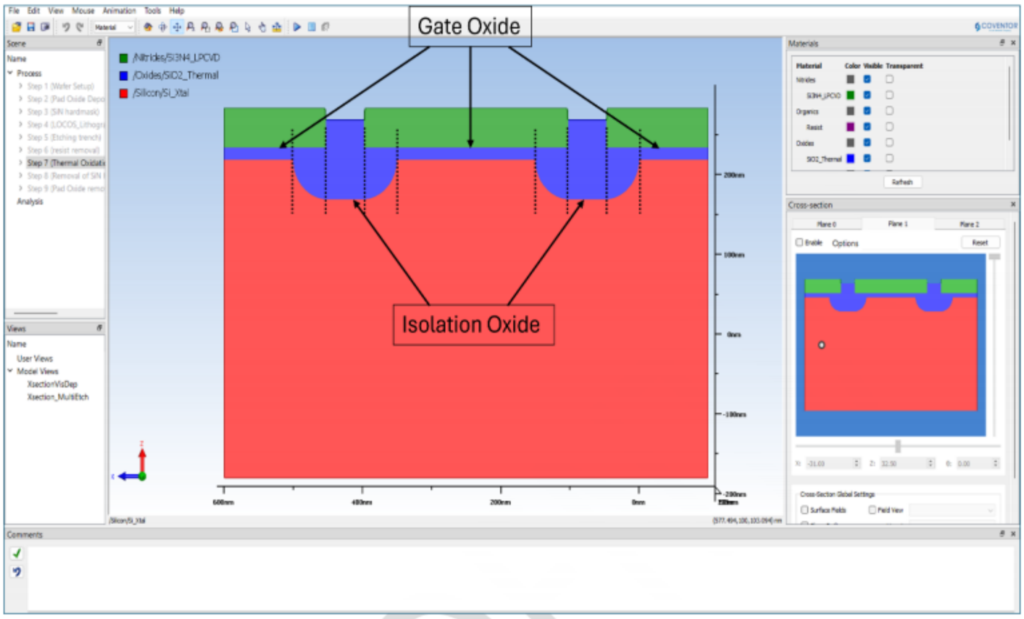
File Edit View Mouse Animation Tools Help
Material
Scene
Name
Process
Step 1 (Wafer Setup)
Step 2 (Pad Oxide Depo
Step 3 (SiN hardmask)
Step 4 (LOCOS_Lithogr
Step 5 (Etching trench)
Step 6 (resist removal)
Step 7 (Thermal Oxidati
Step 8 (Removal of SiN
Step 9 (Pad Oxide remo
Analysis
Views
Name
User Views
Model Views
XsectionVisDep
Xsection_MultiEtch
🟩 /Nitrides/Si3N4_LPCVD
🟦 /Oxides/SiO2_Thermal
🟥 /Silicon/Si_Xtal
Gate Oxide
Isolation Oxide
200nm
100nm
0nm
-100nm
-200nm
600nm
400nm
200nm
0nm
Materials
Material Color Visible Transparent
Nitrides
Si3N4_LPCVD
Organics
Resist
Oxides
SiO2_Thermal
Refresh
Cross-section
Plane 0
Plane 1
Plane 2
Enable Options
Reset
Cross-Section Global Settings
Surface Fields
Field View
Comments
/Silicon/Si_Xtal
COVENTOR

Figure 5 : Simulated Gate and Isolation Oxide

LOCOS takes advantage of the different oxidation rates of silicon and silicon nitride to create isolated regions on a silicon wafer. Silicon nitride acts like a mask, preventing oxide from forming on covered areas. A thin layer of oxide, called the pad oxide, is added between the silicon and nitride to prevent stress during the process.

The final product is an insulating layer (isolation oxide) surrounding the desired regions. However, due to the pad oxide, a slight "bird's beak" extension of oxide forms under the nitride mask. The size of this beak depends on the process duration and the thickness of the involved layers.

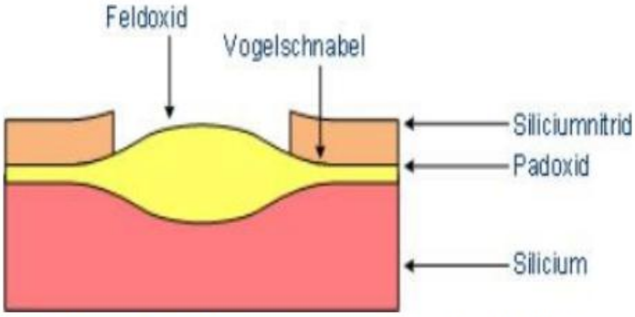
Feldoxid
Vogelschnabel
Siliciumnitrid
Padoxid
Silicium

Figure 6 : Reference image for Bird's Beak

In Figure 7 the area where bird's beak is supposed to formed is marked which should be like as reference image (Figure 6).

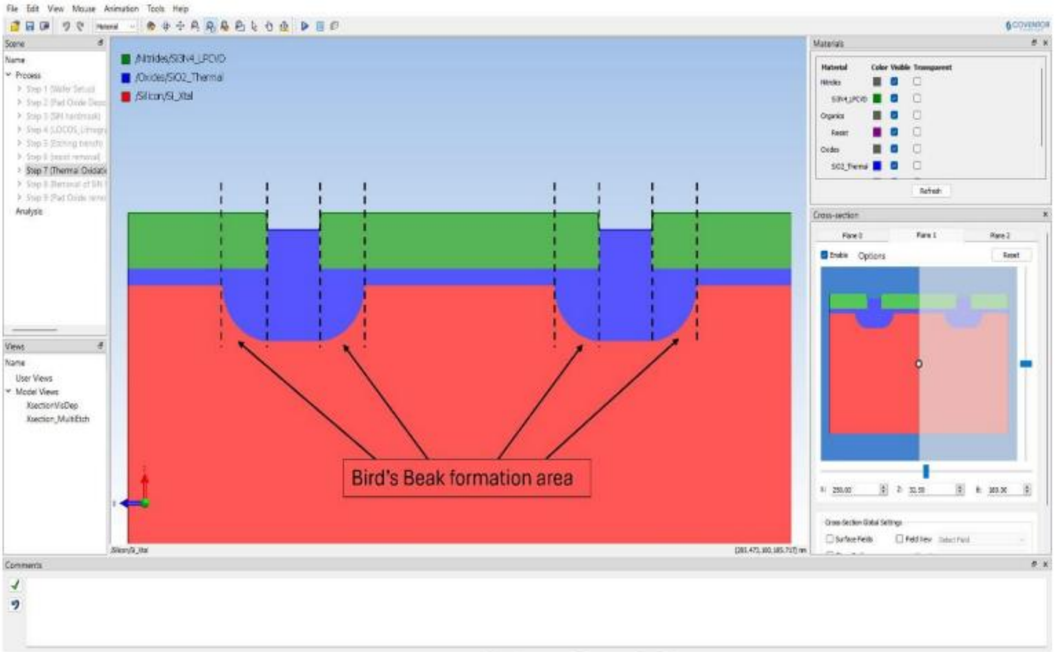

Figure 7 : Simulated Bird's Beak

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As visible in the image, no Bird's beak is formed due to limitations of SEMulator3D software.

LOCOS is a less complex technique for isolation but it has some limitations like,

- **Bird's beak formation** – Due to lateral diffusion of pad oxide during thermaloxidation we observe nitride layer lifting from corners.

- **Scalability problem** – As our devices are becoming smaller the precision needed to be maintained during the process increases, due to lateral oxidation of silicon while growing isolation oxide the active area becomes less than desired.

To overcome these limitations, we can perform it at high temperatures as "When thermal oxidation is performed at high temperatures, less nitride lifting, and therefore a shorter and lower bird's beak is seen. This is due to the increased oxidation rate compared to the diffusion rate. Since the diffusion of oxidant under the nitride layer is the main reason for the bird's beak phenomenon, less diffusion exposure results in decreased bird's beak effects."

Also, we can use other techniques such as Shallow Trench Isolation to perform oxide isolation.

1. Question1(a)

This section deals with finding the optimal active area i.e. where field oxide thickness becomes equal to gate length.

Gate length will be dependent on several parameters like gate pitch, isolation oxide thickness, length of isolation oxide, etc.

Table 2 : Case for finding optimal active area

S.No.	Desired Isolation oxide thickness (nm)	Measured Isolation thickness (depth) (nm)	Measured Gate length (nm)	Measured Isolation length (nm)
1.	100			
2.	120			
3.	150			
4.	180			
5.	200			
6.	300			
7.	320			

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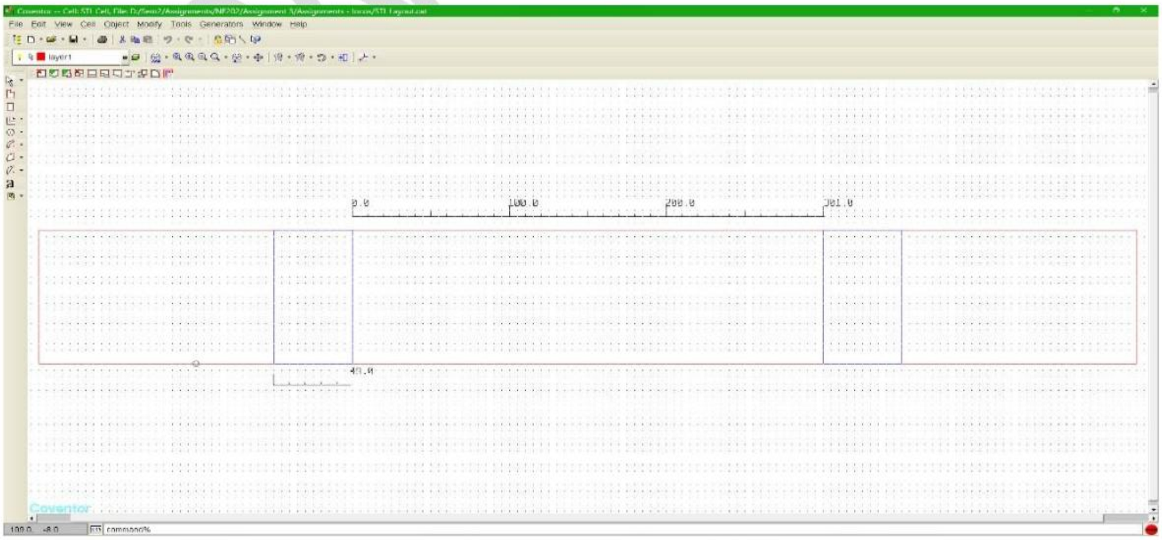
File
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0.0
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40.0
Coventor
command%

Figure 8 : Layout for finding optimal active area

Assignment 3.1.(a)

Study the effect of isolation thickness on the device performance.

1. Perform simulations by varying the isolation thickness over an appropriate range of values.
2. Plot the variation of the relevant device parameters/performance parameters with respect to isolation thickness.
3. Clearly present the simulation results obtained for each variation of isolation thickness.
4. Show the process flow/steps involved in fabricating or designing the device.
5. Provide the cross-sectional view of the device, clearly indicating all important dimensions, materials, and regions.
6. Briefly explain the observed variation in the device parameters with changes in isolation thickness.
7. Measure optimal gate area and isolation oxide thickness.

2. Question1(b)

In this part we analyze the variation in gate length with changes in mask layout. The distance between two trenches (planned gate pitch) in layout is varied from 350nm to 150nm as shown by Table 3,

Table 3 : Cases for Variation of Planned Gate pitch vs gate length

S.No.	Planned gate pitch (nm)	Measured Isolation thickness (depth) (nm)	Measured Gate length (nm)	Measured Isolation length (nm)
1.	350			
2.	300			
3.	250			
4.	200			
5.	150			

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- Isolation oxide thickness entered = 100 nm.

Assignment 3.1.(b)

Study the relationship between the planned gate pitch and gate length for the given case.

1. Perform the required analysis/simulation by varying the gate length over an appropriate range of values.
2. Plot the graph showing the variation of Planned Gate Pitch versus Gate Length.
3. Present and discuss the obtained results, highlighting the relationship between gate length and planned gate pitch.
4. Show the complete process flow/steps involved in the fabrication or implementation of the given device.
5. Provide the cross-sectional view of the device, clearly indicating the relevant layers, materials, dimensions, gate length, and gate pitch.

6. Determine and measure the smallest achievable gate pitch from the obtained analysis, and report the corresponding gate length at which the minimum gate pitch is achieved.